THE UNIVERSITY OF AZAD JAMMU AND KASHMIR, MUZAFFARABAD

Department of Software Engineering

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LAB:	**02**
COURSE TITLE:	**Computer Networks**
COURSE CODE:	**CS-2203**
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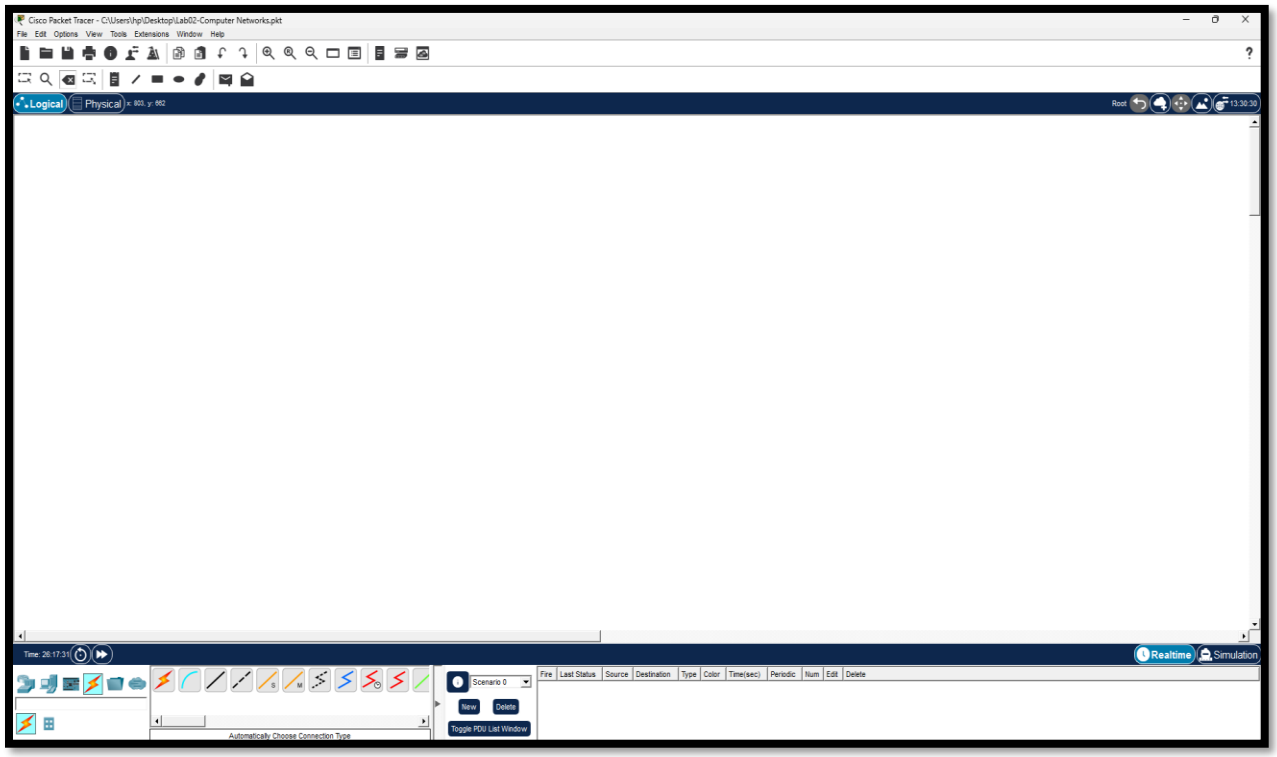
Lab 02: Exploring Basics Concepts of Packet Tracer

Objective

To understand basic networking concepts, identify common networking devices, use Cisco Packet Tracer, and create a simple LAN to test connectivity between devices.

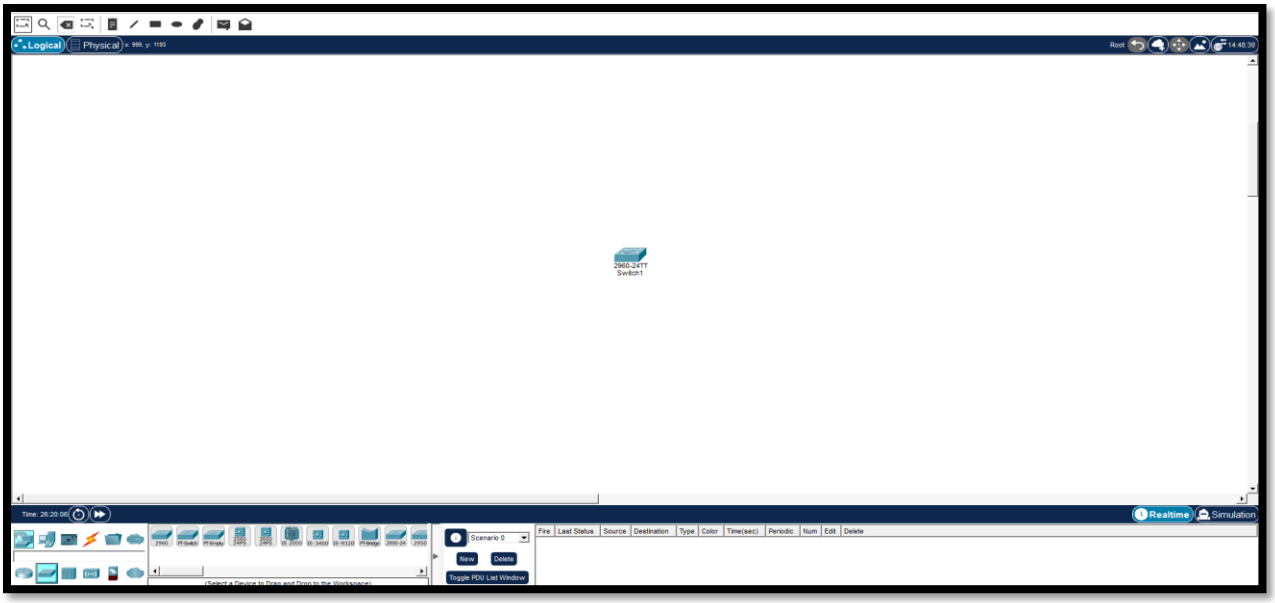
Step 1: Opening Cisco Packet Tracer

Open Cisco Packet Tracer from the system. The main interface appears showing the workspace, device panel, and connection tools.



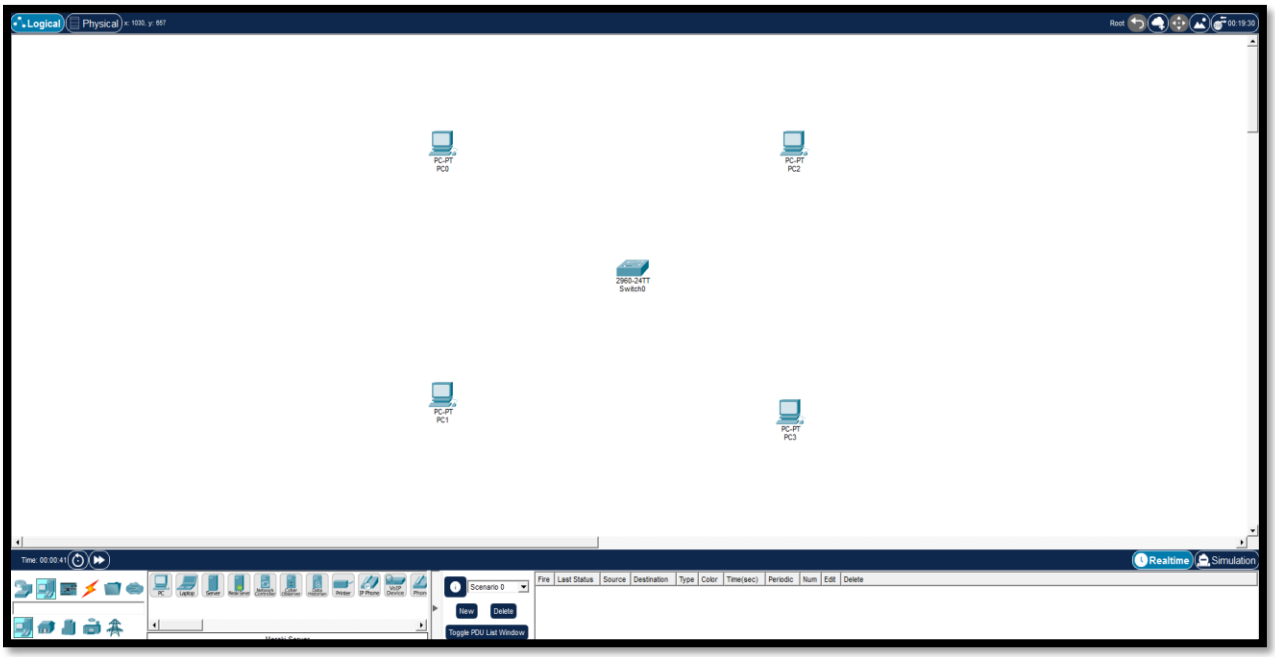
Step 2: Adding a Switch

From **Network Devices → Switches**, drag a **2960 switch** into the workspace. The switch will be used to connect multiple devices.



Step 3: Adding End Devices

From **End Devices**, drag **two PCs (PC0 and PC1…)** into the workspace. These will act as hosts in the network.

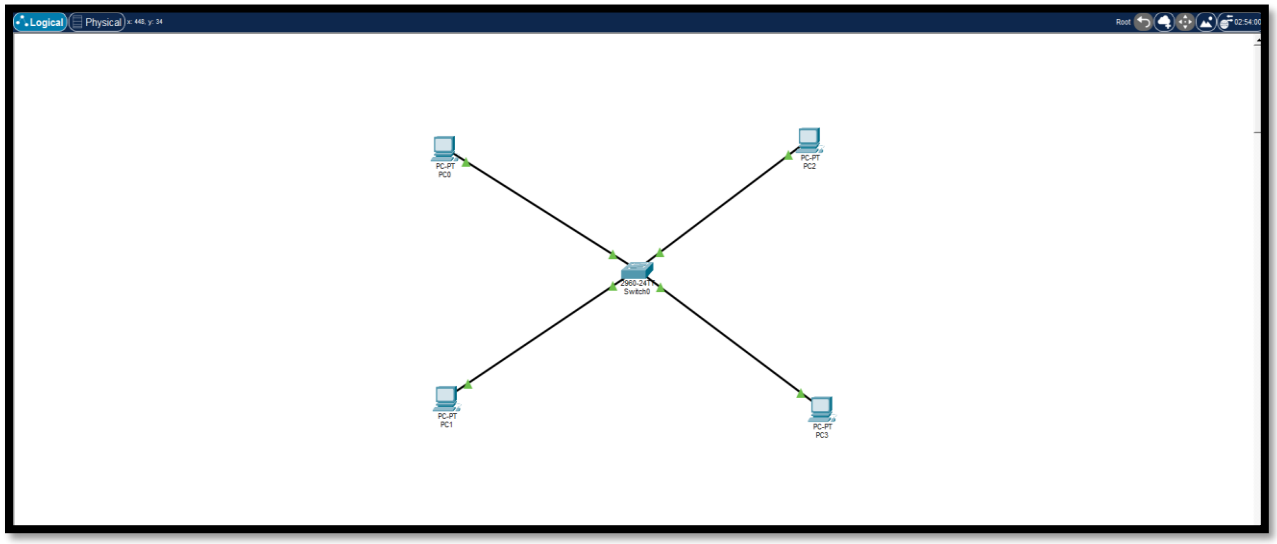


Step 4: Connecting Devices

Select **Automatically choose Connection type** from the connections panel.
Connect:

- **PC0 → Switch**

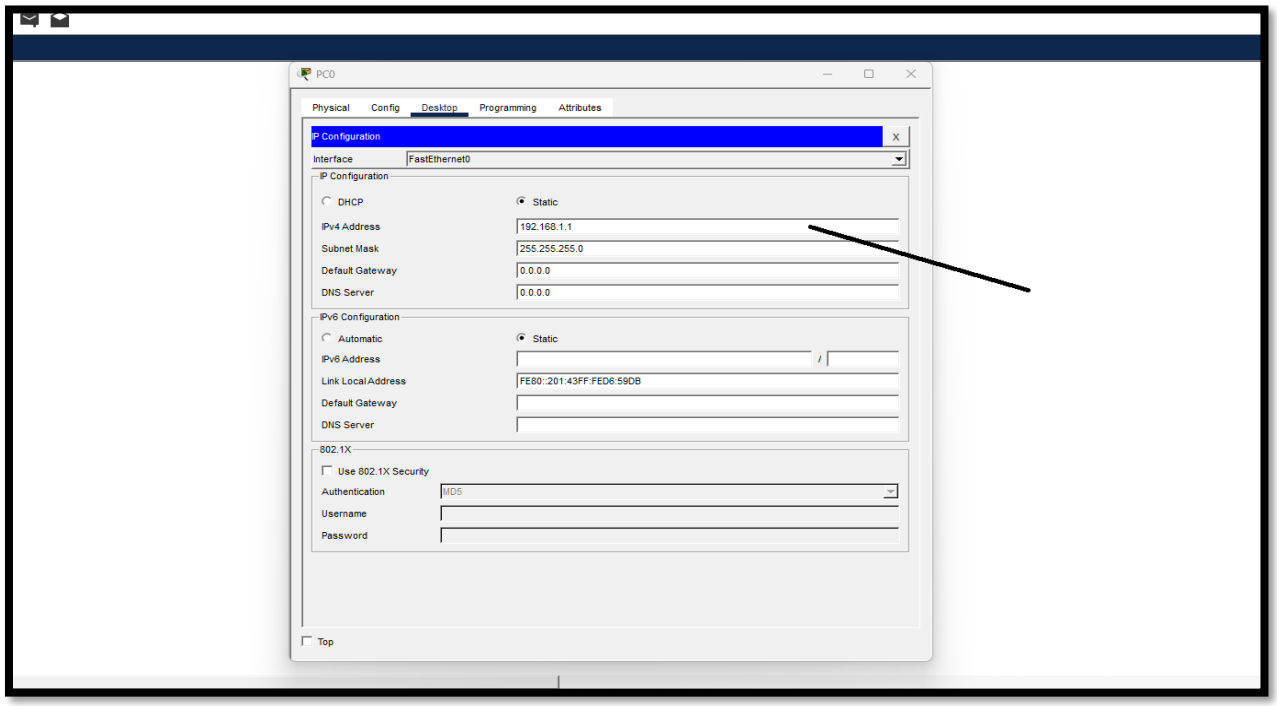
- **PC1 → Switch**



Step 5: Configuring IP Address for All End devices

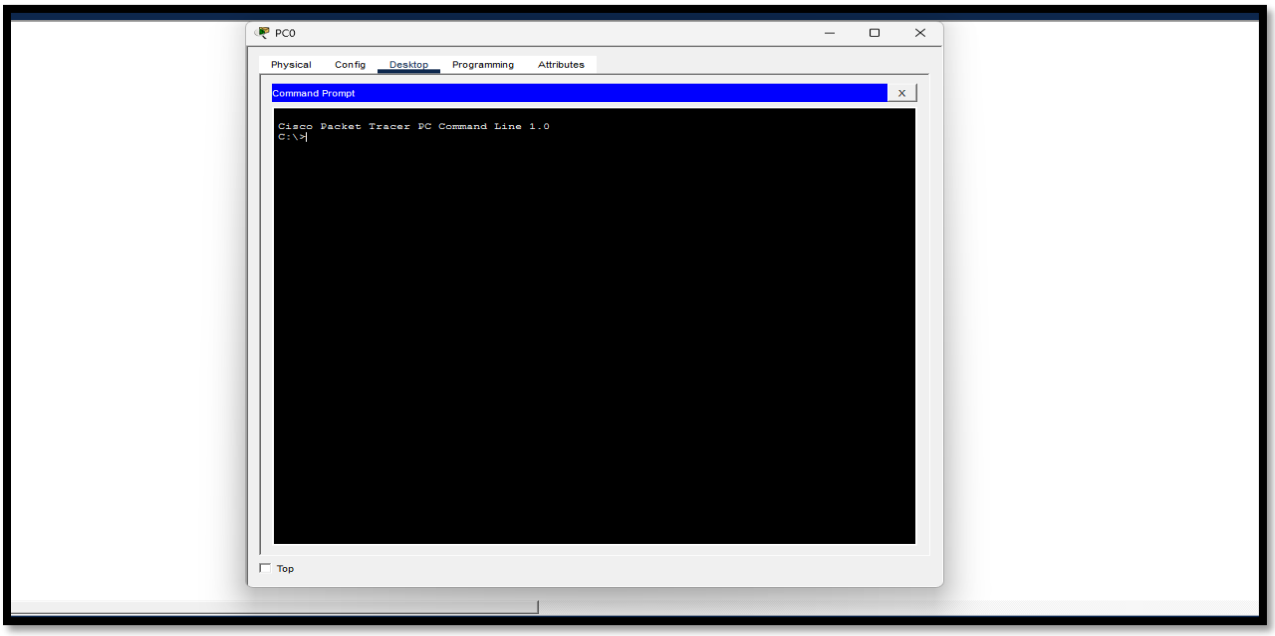
Open PC0 …. P3 → Desktop → IP Configuration.

IP Address: **192.168.1.1 …**
Subnet Mask: **255.255.255.0**



Step 7: Opening Command Prompt

Open **PC0 → Desktop → Command Prompt** to run network commands.



Step 8: Running ipconfig

Run:

ipconfig

This command displays the IP address and subnet mask configuration.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:43FF:FED6:59DB
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.1
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>
```

Step 9: Testing Connectivity using ping

Run:

ping 192.168.1.2

Successful replies confirm communication between the PCs.

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>
```

Step 10: Running tracert

Run:

tracert 192.168.1.2

This shows the path packets take to reach the destination.

```
C:\>tracert 192.168.1.3

Tracing route to 192.168.1.3 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.3

Trace complete.

C:\>
```

Step 11: Viewing ARP Table

Run:

arp -a

This command shows the mapping between IP addresses and MAC addresses.

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.2           000a.f3d4.83c4        dynamic
  192.168.1.3           00d0.ba87.70b7        dynamic

C:\>
```

Step 12: Running netstat

Run:

netstat

This command displays network statistics and active connections.

```
C:\>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
C:\>
```

Conclusion

A simple LAN was created using Cisco Packet Tracer by connecting two PCs through a switch. IP addresses were configured, and commands such as ipconfig, ping, tracert, arp, and netstat were used to verify connectivity and analyze the network.